

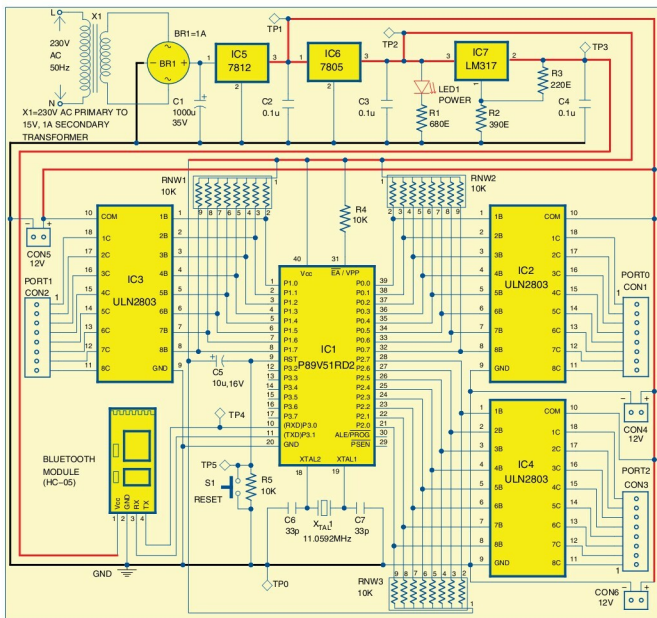


Fig. 2: Circuit of the 'Home Automation through Bluetooth' application

CON4 through CON6. 5V from the output of IC6 is given to microcontroller IC1 and adjustable regulator IC7. 3.3V output from IC7 is required to operate the Bluetooth module. The glowing of LED1 indicates presence of power in the circuit.

Microcontroller. The P89V51RD2 is an 8-bit 80C51 microcontroller with 64kB Flash and 1024 bytes of data RAM. It has three 16-bit timers/counters, eight interrupt sources with four priority levels, serial peripheral interface (SPI) and enhanced UART, programmable counter array with

PWM and capture/compare functions, and four 8-bit input/output (I/O) ports. It also has an on-chip oscillator and clock circuitry, which is operated

TABLE II
Test Points

Test point	Details
TP0	0V, GND
TP1	12V
TP2	5V
TP3	3.3V
TP4	Train of pulses
TP5	Low when S1 is pressed

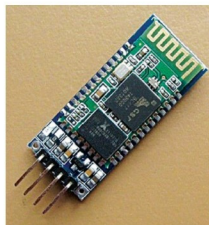


Fig. 3: HC-05 Bluetooth module